Lesson 12.2 - The Chi Square Distribution Test of a Single Standard Deviation

> **Video:** Chi Square Distribution Test of a Single Variance or Standard Deviation by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

Test statistic

To test a claim about a population **variance** σ^2 (or standard deviation σ):

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}, \qquad \mathrm{df} = n - 1$$

- s^2 = sample variance
- σ_0^2 = the variance claimed in H_0

H_a	Test	Rejection region
$\sigma^2 < \sigma_0^2$	Left-tailed	$\chi^2 <$ critical value, with area α on the left (look up right-tail area $1-\alpha$)
$\sigma^2 > \sigma_0^2$	Right-tailed	$\chi^2 >$ critical value, with right-tail area α
$\sigma^2 \neq \sigma_0^2$	Two-tailed	both tails, $\alpha/2$ each

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Example - is the spread of exam scores smaller?

The math department's exam scores have $\sigma = 8.6$. A professor thinks it is **less**. A sample of 20 scores has $s = 6.9$. Test at $\alpha = 0.05$.

Known values: $\sigma_0^2 = 8.6^2,\ \ s^2 = 6.9^2,\ \ n = 20,\ \ \mathrm{df} = 19$

Hypotheses:

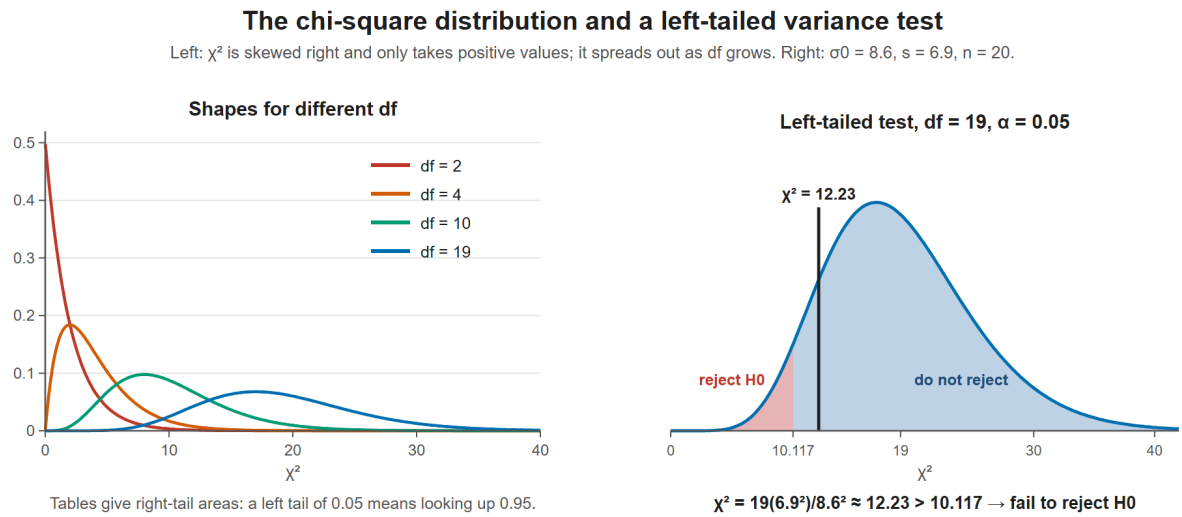
$$H_0 : \sigma^2 = 8.6^2 \qquad H_a : \sigma^2 < 8.6^2 \quad (\text{left-tailed})$$

Critical value: the left-tail area is 0.05, so the right-tail area is 0.95. With df = 19 the table gives $\chi^2 = 10.117$. Reject H_0 only if the calculated χ^2 is **less than 10.117**.

Test statistic:

$$\chi^2 = \frac{19 \times 6.9^2}{8.6^2} = \frac{19 \times 47.61}{73.96} \approx 12.23$$

$12.23 > 10.117$, which is in the do-not-reject region. **Fail to reject H_0.** At the 5% level the sample does not provide enough evidence that the standard deviation is less than 8.6.



Left: χ^2 is never negative and is skewed to the right; with more degrees of freedom it moves right and spreads out. Right: for the left-tailed variance test the rejection region is the left tail below 10.117, and 12.23 does not reach it.

Key takeaways

- Testing a variance or SD: $\chi^2 = (n-1)s^2/\sigma_0^2$ with df $= n - 1$.
- χ^2 tables list **right-tail areas**. For a left-tailed test, look up $1 - \alpha$.
- Left-tailed: reject only when χ^2 is **below** the critical value.